

UNIT I-CRYSTAL STRUCTURE AND LATTICE DEFECTS

1.1 INTRODUCTION

Solids are brittle, some are ductile, some are malleable, some are strong, some are weak, some are good conductors of heat and electricity, some are bad conductors of heat and electricity, some are magnetic, some are non-magnetic and so on, but all these materials are composed of atoms and molecules.

The constituent particles, i.e., atoms (or) molecules of matter are held together by the forces of attraction. The attractive forces which hold the constituent particles of a substance together are called **bonds**.

The reason for these different properties of the solids is due to their structure. i.e., the behaviour of the solid materials is closely related to the structure of the material.

1.2 CLASSIFICATION OF SOLIDS

Based on the internal atomic structure, the solids can be classified into two categories namely **(i) Crystalline and (ii) Non-crystalline (or) Amorphous solids**.

1.3 CRYSTALLINE SOLIDS (OR) CRYSTALS

Crystalline solids (or) crystals are those in which the constituent atoms (or) molecules are arranged in an orderly fashion throughout, in a three dimensional pattern.

Each atom (or) molecule is fixed at a definite point in space, at a definite distance from each other and in a definite angular orientation to all other atoms (or) molecules surrounding it. Therefore, crystalline solids have well defined geometrical form (pattern) i.e., defined faces and angles between them.

Further when crystal breaks, all the broken pieces will have a regular shape. *The crystalline solids have directional properties and therefore they are called **isotropic substances**.*

A crystalline material will be in two forms viz,

- (i) **Single crystal**, in which the solid contains only one crystal. These single crystals are produced artificially from their vapour (or) liquid state.
- (ii) **Poly crystal**, which has an aggregate of many small crystals (or) grains separated by well defined grain boundaries. These crystals will have a sharp melting point.

Examples for crystalline solids: Diamond, Copper, Platinum, Silver, Polonium, Gold, Aluminium, Molybdenum, Nickel, Cadmium, Iron etc.

Crystallography: *The study of the geometric form and other physical properties of crystalline solids, using X-rays (or) electron beam (or) neutron beam etc., is termed as the science of crystallography.*

1.4 AMORPHOUS SOLIDS (OR) NON-CRYSTALLINE SOLIDS

“Amorphous” means “without form”

In amorphous solids the constituent particles i.e., atoms (or) molecules are not arranged in an orderly fashion. In other way we can say that in amorphous solids the same atomic groups are arranged randomly in all directions.

*These solids have no directional properties and therefore they are called **isotropic substances**. These amorphous solids do not have a sharp melting point.*

Examples for amorphous solids: Plastics, Rubber and Glass.

1.5 DIFFERNECE BETWEEN CRYSTALLINE AND NON-CRYSTALLINE MATERIAL

S.No.	Crystalline material	Non-crystalline material
1	They have definite and regular geometrical shapes which extend throughout the crystal.	They do not have definite geometrical shape.
2	They are anisotropic.	They are isotropic.
3	They are most stable.	They are less stable.
4	They have sharp melting point.	They do not have sharp melting point.
5	Examples: Diamond, NaCl, KCl, Copper, Iron, etc.	Examples: Glasses, Plastics, Rubber etc.

1.6 FUNDAMENTALS OF CRYSTALS AND ITS STRUCTURE

1.Lattice

Lattice is defined as an array of points which are imaginarily kept to represent the position of atoms in the crystal such that every lattice point has got the same environment as that of the other and hence one lattice point cannot be distinguished from the other lattice point. It is an imaginary concept.

2.Space lattice (or) crystal lattice

A three dimensional collection of points in space are called **space lattice (or)crystal lattice**. The environment about any particular point is in every way the same.

(or)

A geometrical representation of the crystal structure in terms of lattice points are called **space lattice**, provided the environment about every point is identical to that of every other point.

3.Lattice points

Lattice points denote the position of atoms (or) molecules in the crystal Fig.1.1 (a)

(or)

*The points in the space lattice are called **lattice points**.*

4. Basis (or) Motif

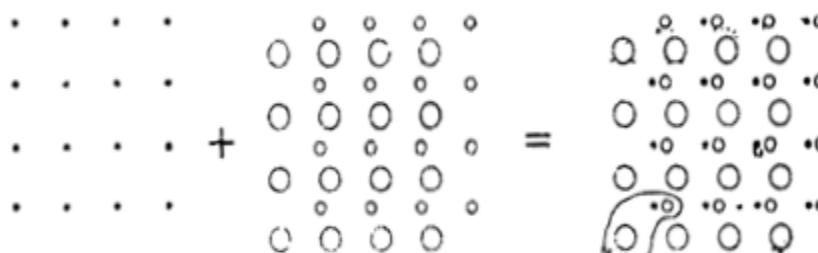


Fig. 1.1

Fig. 1.1 b

Fig. 1.1 c

Basis (or) Motif is an unit assembly of atoms (or) molecules which are identified with respect to the position of lattice points, identical in composition arrangement and orientation as shown in Fig.1.1 (b)

In NaCl and KCl the basis is two atoms and in CaF₂ the basis is three atoms. But there are structures for which the basis can have more than 1000 atoms.

5. Crystal structure

The crystal structure is formed by the addition of basis [Fig.1.1(b)] to every lattice point of the lattice [Fig.1.1(a)].

The logical relation for crystal structure is as shown in Fig.1.1(a), (b) and (c).

i.e., Fig.(1.1 a)+Fig. (1.1 b)=Fig. (1.1 c)

i.e., Space lattice + Basis = Crystal structure

6. Lattice planes

*A set of parallel and equally spaced planes in a space lattice, which are formed with respect to the lattice points are called **lattice planes**.*

The different ways of drawing the lattice planes for the array of lattice points is shown in Fig.1.2



Fig 1.2 lattice planes

7. Lattice parameters (or) Unit cell parameters

*The lines drawn parallel to the lines of intersection of any three faces of the unit cell which do not lie in the same plane are called **crystallographic axes**.*

The intercepts a, b and c are nothing but the edges of the unit cell, (i.e., the distance between two lattice points) which defines the dimensions of an unit cell. These intercepts are known as its **primitives (or) characteristic intercepts** on the axes.

These three quantities a, b and c are also called the fundamental translational vectors (or) axial lengths. The angles between (a, b) , (b, c) and (c, a) are denoted by γ, α and β respectively. These three angles (α, β and γ) are called **interfacial angles**.

Both, the intercepts (a, b, c) and interfacial angles (α, β, γ) constitutes the lattice parameters (or) cell parameters of the unit cell.

The actual size of the unit cell be determined by knowing the values of intercepts and interfacial angles.

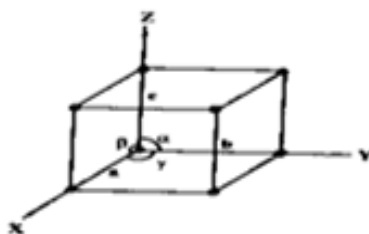


Fig 1.3 interfacial angles

7. Unit cell

The unit cell is defined as the smallest geometric Figure, the translational repetition of which in all over the three dimensions gives the actual crystal structure (**or**) The unit cell may also be

defined as the fundamental elementary pattern with minimum number of atoms, molecules (or) groups of molecules which represents the total characteristics of the crystal. Fig. 1.4



Fig 1.4 unit cell

9. Primitive cell

A primitive cell is the simplest type of unit cell which contains only one lattice point per unit cell (contains lattice points at its corner only) Example: Simple Cubic (SC)

10. Non-primitive Cell

If there is more than one lattice point in a unit cell, it is called a non-primitive cell.

Example: BCC and FCC contain more than one lattice point per unit cell. If the number of lattice points per unit cell is two (BCC), three and four (FCC), then the unit cell is called **doubly primitive, triply primitive** and quadruply primitive respectively.

1.7 THE CRYSTAL SYSTEMS

Crystals are classified into 7 crystal systems on the basis of lattice parameters viz, (i) axial lengths a, b, c and (ii) interfacial (axial) angles α, β, γ .

The 7 basic crystal systems are

1. Cubic. 2. Tetragonal 3. Orthorhombic 4. Trigonal (or) Rhombohedral 5. Hexagonal 6. Monoclinic and 7. Triclinic

The 7 crystal systems are discussed briefly one by one as follows

1. Cubic crystal system

In cubic crystal system, the three crystal axes are perpendicular to each other and the axial lengths are same in three axes as shown in Fig. 1.5.

$$\text{Lattice parameters are } \left[\begin{array}{l} a = b = c \\ \alpha = \beta = \gamma = 90^\circ \end{array} \right]$$

Cubic lattices may be in any one of these three forms viz, Simple Cubic (SC) (or) Body-Centered Cubic (BCC) (or) Face Centered Cubic (FCC).

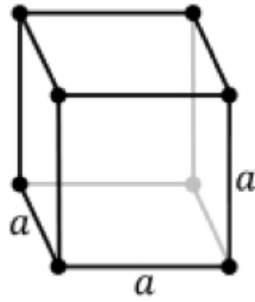


Fig 1.5 Cubic crystal

2. Tetragonal crystal system

In tetragonal system, the three crystal axes are perpendicular to each other. Two of the three axial lengths are the same, but the third axial length is different, as shown in Fig. 1.6.

$$\text{Lattice parameters are } \begin{bmatrix} a = b \neq c \\ \alpha = \beta = \gamma = 90^\circ \end{bmatrix}$$

Tetragonal lattices may be in any one of these two forms viz., Simple (or) Body-centered.

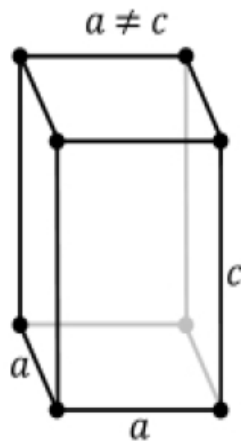


Fig 1.6 Tetragonal crystal

3. Orthorhombic crystal system

In orthorhombic crystal system, the three Crystal axes are perpendicular to each other. In this system, all the three axial lengths are of unequal lengths (different) shown in Fig.1.7

$$\text{Lattice parameters are } \begin{bmatrix} a \neq b \neq c \\ \alpha = \beta = \gamma = 90^\circ \end{bmatrix}$$

Orthorhombic lattice may be in any one of these four forms i.e., simple, (or) base centered (or) body-centered (or) face centered.

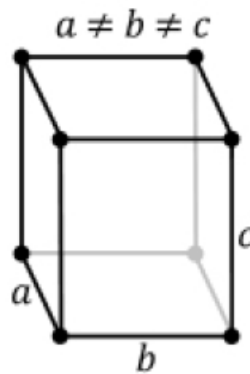


Fig 1.7 Orthorhombic crystal

4. Trigonal (or) Rhombohedral crystal system

In trigonal crystal system, the three axes are inclined to each other at an angle other than 90° . The three axial lengths are equal along the axes, as shown in Fig. 1.8.

Lattice parameters are $\left[\begin{array}{l} a = b = c \\ \alpha = \beta = \gamma \neq 90^\circ \end{array} \right]$

Trigonal lattice has only one form i.e., simple.

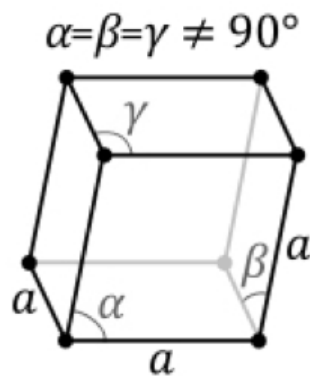


Fig 1.8 Trigonal (or) Rhombohedral crystal

5. Hexagonal crystal system

In hexagonal crystal system, two of the crystal axes are 120° apart, while the third is perpendicular to both of them.

The axial lengths are the same along the axes that are 120° apart, but the axial length along the third axis is different, as shown in Fig. 1.9.

Lattice parameters are $\left[\begin{array}{l} a = b \neq c \\ \alpha = \beta = 90^\circ \text{ and } \gamma = 120^\circ \end{array} \right]$

Hexagonal lattice has only one form i.e., Simple.

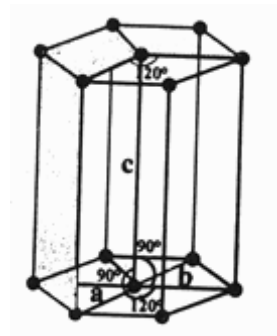


Fig 1.9 Hexagonal crystal

6. Monoclinic crystal system

In monoclinic crystal system, two of the crystal axes are perpendicular to each other, but the third is obliquely inclined. The three axial lengths are different along the axes as shown in Fig.1.10.

Lattice parameters are $\left[\begin{array}{l} a \neq b \neq c \\ \alpha = \gamma = 90^\circ \neq \beta \end{array} \right]$

Monoclinic lattices may be in any one of these two forms viz. simple (or) base centred.

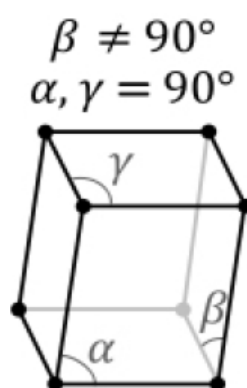


Fig 1.10 Monoclinic crystal

7. Triclinic crystal system

In triclinic crystal system, all the three crystal axes are not perpendicular to each other. The axial lengths are also not equal (different) along the three axes, as shown in Fig.1.11

Lattice parameters are $\begin{bmatrix} a \neq b \neq c \\ \alpha \neq \beta \neq \gamma \neq 90^\circ \end{bmatrix}$

Triclinic lattice has only one form i.e., Simple.

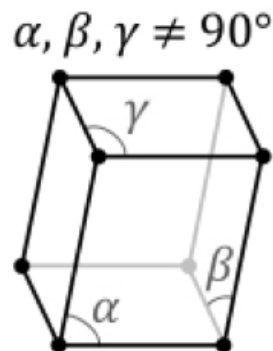


Fig 1.11 Triclinic crystal

1.8 BRAVAIS LATTICES

Bravais in the year 1880 showed that there are 14 possible types of space lattices in the 7 crystal systems.

*According to Bravais, there are only 14 possible ways of arranging points in space lattice from the 7 crystal systems such that, all the lattice points have exactly the same surroundings. These 14 space lattices are called the **Bravais lattices**.*

The 14 possible Bravais lattices drawn from the 7 crystal systems are shown in Fig.1.12 below

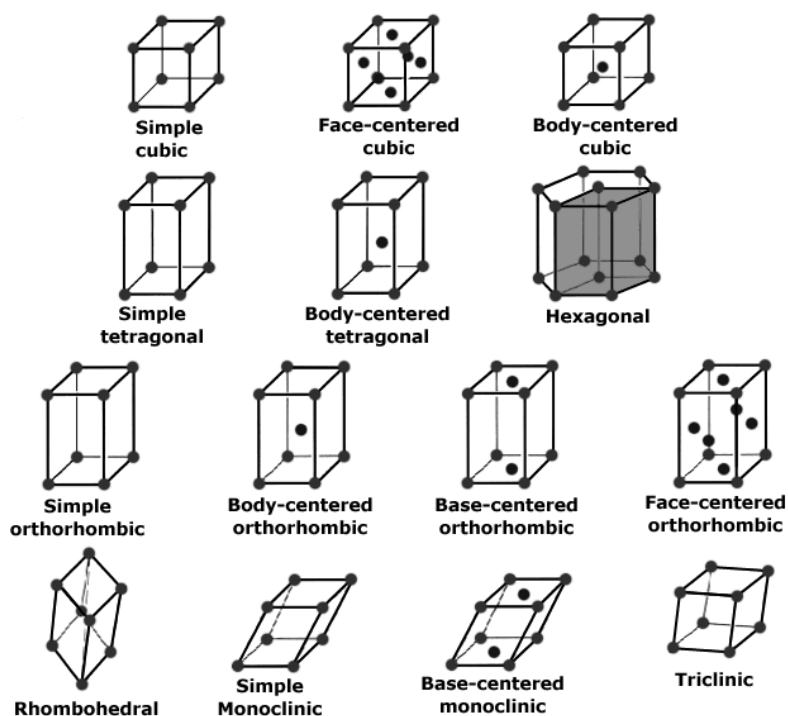


Fig 1.12

Table 1.2 shows the seven crystal systems, which are divided into 14 Bravais lattices along with the unit cell parameters and examples.

Table 1.2

The Seven Crystals Systems, divided into 14 Bravais lattices				
S. No	Crystal System	Bravais lattice	Unit cell Parameters	Examples
1.	Cubic	Simple Body-centered Face-centered	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	NaCl, CaF ₂ , NaClO ₃
2.	Tetragonal	Simple Body-centered	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	NiSO ₄ , SnO ₂ , Indium, white tin
3.	Orthorhombic	Simple Base-centered Body-centered Face-centered	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	KNO ₃ , BaSO ₄ , MgSO ₄ Sulphur, Topaz
4.	Trigonal (Rhombohedral)	Simple	$a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	CaSO ₄ , Bi, Sb, Calcite
5.	Hexagonal	Simple	$a = b \neq c$ $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$	Tourmaline, Quartz
6.	Monoclinic	Simple Base-centered	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ \neq \beta$	Na ₂ SO ₄ , FeSO ₄ NO ₂ SO ₃
7.	Triclinic	Simple	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	CuSO ₄ , K ₂ Cr ₂ O ₇

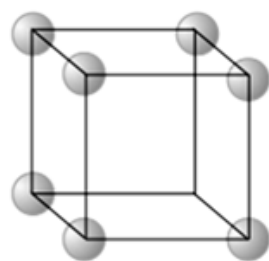
1.9 STRUCTURE OF CUBIC AND HEXAGONAL SYSTEM

Cubic system

The cubic structure is the simplest type of array in which, the atoms can take positions at the corners, faces and also at the centre of the cube. There are three types of lattices possible in this system depending on the positions of the lattice points (atoms) in the unit cell, viz.,

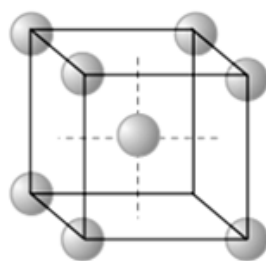
1. Simple Cubic (SC) (or) Primitive (P)
2. Body Centered Cubic (BCC) and
3. Face Centered Cubic (FCC)

1. In **simple cubic lattice**, there is one atom at each of the eight corners of the unit cell, as shown in the Fig. (1.13) Example: Polonium
2. In a **body centered cubic** (BCC) lattice the atoms are located at all the eight corners and in addition one more atom is located at the centre of the unit cell, so called the body centered atom, as shown in Fig. (1.14) Examples: Tungsten, Chromium, Molybdenum, etc.,
3. In a **face centered cubic** (FCC) **lattice** the atoms are located at all the eight corners and in addition the atoms are located in all the six faces in a fashion of one in each face of the unit cell as shown in Fig.1.15 Examples: Copper, Al, Ni, Au, Pb, etc.



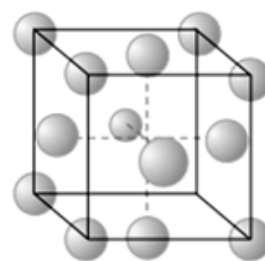
simple cubic

Fig 1.13



body-centered cubic

Fig 1.14

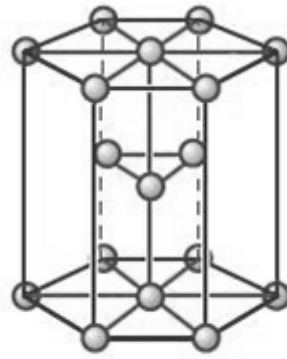


face-centered cubic

Fig 1.15

Hexagonal System

In Hexagonal close packed structure; atoms are located at each of the twelve corners of the hexagon and 2 atoms, one at the centre of top Hexagonal face and another at the centre of bottom hexagonal face. Along with these atoms there are 3 more atoms in the middle layer, situated in between the top and bottom hexagonal face of the cell, in alternative vertical faces as shown in Fig.1.16. Examples: Magnesium, Osmium, Cobalt, titanium, etc.,



hexagonal close-packed (hcp)

Fig 1.16

1.10 PARAMETERS DETERMINING THE CRYSTAL STRUCTURE OF MATERIALS

Let us discuss some of the important parameters which are used to describe the crystal structure.

1. Number of atoms per unit cell (or) Effective number

The total number of atoms present in (or) shared by an unit cell is known as number of atoms per unit cell.

2. Atomic radius

Atomic radius is defined as half of the distance between any two nearest neighbour atoms which have direct contact with each other, in a crystal of a pure element.

It is usually expressed in terms of cube edge a .

3. Co-ordination number

Co-ordination number is the number of nearest neighboring atoms to a particular atom.

(or)

Co-ordination number is the number of nearest neighbouring directly surrounding a given atom.

4. Atomic packing factor (or) packing density (or) density of packing

Atomic packing factor is defined as the ratio between the volumes occupied by the total number of atoms per unit cell (v) to the total volume of the unit cell (V).

$$\text{i.e., APF} = \frac{\text{Total number of atoms per unit cell} \times \text{Volume of one atom}}{\text{Total volume of the unit cell}} = \frac{(v)}{(V)}$$

Let us discuss all the above parameters one by one for a simple cubic structure.

1.11 CRYSTAL STRUCTURES- FACE CENTERED CUBIC STRUCTURE (FCC)

Number of atoms per unit cell:

The total number of atoms present in (or) shared by an unit cell is known as number of atoms per unit cell.

In this case, we have two types of atoms viz,

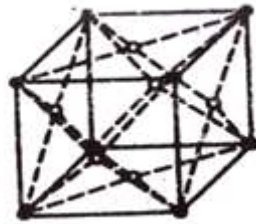


Fig 1.17

(i) Corner atoms (ii) face centered atoms. There are 8 corner atoms, one at each corner of the unit cell and six atoms at the centers of six faces of unit cell as shown in Fig. (1.17)

(i) Number of corner atoms per unit cell.

Each and every corner atoms is shared by 8 adjacent unit cells.

$$\therefore \text{The total number of corner atoms per unit cell} = \frac{1}{8} \times 8 = 1$$

(ii) Number of face centered atoms per unit cell.

Each face centered atom is shared by only two unit cells, which lie on either side of the atom.

Similarly we have six face centered atoms in a unit cell.

$$\therefore \text{The total number of face centered atoms per unit cell} = \frac{1}{2} \times 6 = 3$$

$$\left\{ \begin{array}{l} \therefore \text{Total number of atoms} \\ \text{per unit cell in FCC} \end{array} \right. = \left\{ \begin{array}{l} \text{Total number of corner atoms} + \\ \text{Total number of face centered atoms} \end{array} \right\}$$
$$= 1 + 3 = 4$$

Atomic radius

Atomic radius is defined as half of the distance between any two nearest neighbour atoms which have direct contact with each other, in a crystal of a pure element.

It is usually expressed in terms of cube edge a .

In FCC structure, the corner atoms do not touch each other. But each corner atom touches the face centered atoms along the diagonal of the face of the cube. Therefore, the two corner atoms (A and C)

situated at the opposite ends of the same face can be joined by drawing a diagonal as shown in Fig. 1.18.

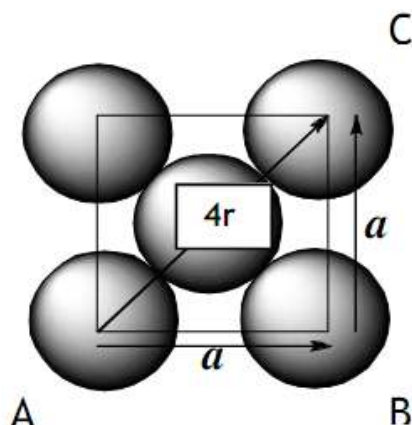


Fig 1.18

From the geometry of the Fig. (b), we can write

$$(AC)^2 = (AB)^2 + (BC)^2$$

$$= a^2 + a^2$$

$$\text{(or)} (AC)^2 = 2a^2$$

$$\therefore AC = a\sqrt{2} \dots\dots\dots(1)$$

but from Fig. (b), the diagonal of the cube $AC = 4r$ (2)

$\therefore$ From equations (1) and (2) we can write $4r = a\sqrt{2}$

$$\therefore \text{Atomic radius } r = \frac{a\sqrt{2}}{4}$$

Co-ordination number:

Co-ordination number is the number of nearest neighbouring atoms to a particular atom.

(or)

Co-ordination number is the number of nearest neighbouring directly surrounding a given atom.

In this system, there are 8 corner atoms, and six face centred atoms, one at the centre of each face.

To calculate the coordination number, for FCC, let us consider a corner atom (X) as shown in Fig. 1.19

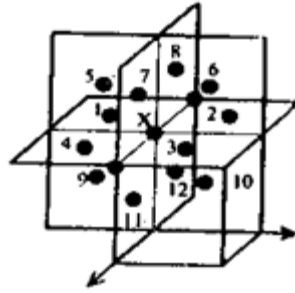


Fig 1.19

In its own plane it has 4 face centered atoms (1,2,3,4) as nearest neighbours. And, similarly in a plane which lie just below this corner atom it has 4 more face centered atoms, (9,10,11,12) as nearest neighbours.

The total number of nearest atoms to any corner atom is $4+4+4=12$.

Hence, the coordination number is 12.

Atomic packing factor:

Atomic packing factor is defined as the ratio between the volumes occupied by the total number of atoms per unit cell (v) to the total volume of the unit cell (V).

$$\text{i.e., } APF = \frac{\text{Total number of atoms per unit cell} \times \text{Volume of one atom}}{\text{Total volume of the unit cell}} = \frac{(v)}{(V)}$$

The number atoms per unit cell = 4

$$\therefore \text{Volume of 4 atoms} = 4 \times \frac{4}{3} \pi r^3$$

We know the radius of atom in FCC is $r = \frac{a\sqrt{2}}{4}$

$$\therefore \text{Volume occupied by the atoms per unit cell (v)} = \frac{16\pi}{3} \left[\frac{a\sqrt{2}}{4} \right]^3$$

$$(\text{or}) \quad (v) = \frac{16\pi}{3} \left[\frac{a^3 2\sqrt{2}}{4 \times 4 \times 4} \right]$$

$$\therefore \text{Volume occupied by the atoms per unit cell (v)} = \pi a^3 \frac{\sqrt{2}}{6}$$

$$\therefore \text{Volume of the unit cell for cubic system (v)} = a^3$$

$$\therefore \text{Atomic packing factor (APF)} = \frac{\pi a^3 \frac{\sqrt{2}}{6}}{a^3}$$

$$(\text{or}) \quad APF = \pi \frac{\sqrt{2}}{6} = 0.74$$

$$\therefore \quad APF = 0.74$$

Therefore, we can say that 74% volume of the unit cell of FCC is occupied by atoms and remaining 26% volume is vacant.

Thus, the packing density is 74 %

Since the packing density is very high, the FCC structure has closely (or) tightly packed structure.

1.12 CRYSTAL STRUCTURES –HCP

In the HCP structure an unit cell contains three types of atoms viz.,

- (i) 12 corner atoms, one at each and every corner of the hexagon, as shown in Fig.1.20
- (ii) 2 base centered atom, one at the top face of the hexagon and another at the bottom face of the hexagon as shown in Fig.1.20.
- (iii) In addition to corner and base atoms, 3 atoms are situated in between the top and bottom face of hexagon, in alternate vertical faces. Also note that these atoms are situated inside the faces, so that they can't be shared by other unit cells, as shown in Fig.1.20

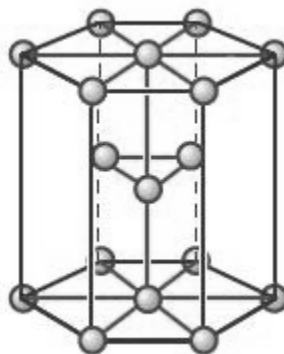


Fig 1.20

Thus it is obvious that each unit cell shares 14 atoms (12 corner atoms + 2 base atoms) with other unit cells and contains 3 atoms within that unit cell.

The common examples of this type of structure are Magnesium, Zinc, Zirconium, Cadmium etc.,

The calculation of 1. Coordination number, 2. Number of atoms per unit cell, 3. Atomic radius and 4. Packing factor is explained below.

Coordination number

Co-ordination number is the number of nearest neighbouring atoms to a particular atom.

In the top and bottom layers, the base centred atom is surrounded by six corner atoms. In the middle layer we have 3 atoms stacked inside the unit cell as shown in Fig.1.21

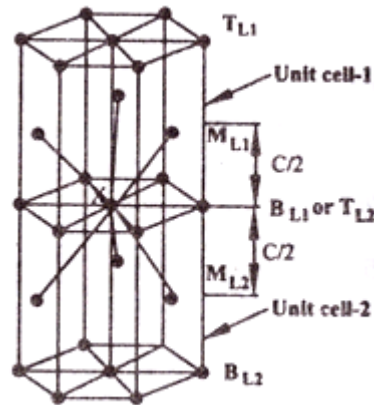


Fig 1.21

Let us consider two unit cells as shown in Fig. 1.21. Let 'x' be the reference atom taken in the bottom layer (BL_1) of unit cell-1 (or top layer [TL_2] of unit cell 2). This atom has a neighbouring atom in its own plane. Further at a distance of $c/2$ it has 3 atoms in the middle layer (ML_1) of unit cell-1 and 3 more atoms in the middle layer (ML_2) of unit cell-2.

Therefore, the total numbers of neighbouring atoms are $6+3+3=12$.

Thus the coordination number is 12.

Number of atoms per unit cell:

The total number of atoms present in (or) shared by an unit cell is known as number of atoms per unit cell.

We know in HCP, there are 3 types of atoms viz., (i) corner atoms
(ii) Base atoms and (iii) middle layer atoms as shown in Fig 1.22.

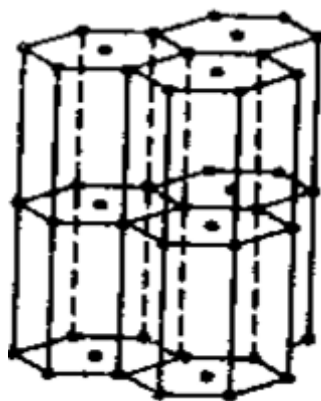


Fig 1.22

(i) The number of corner atoms per unit cell = $\frac{1}{6} \times 12 = 2$

(ii) The number of base centered atoms per unit cell = $\frac{1}{2} \times 2 = 1$

(iii) The total number of middle layer atoms per unit cell = 3

$$\therefore N = 2 + 1 + 3 = 6$$

$\therefore$ Number of atoms per unit cell in HCP structure = 6

Atomic radius

Atomic radius is defined as half of the distance between any two nearest neighbour atoms which have direct contact with each other, in a crystal of a pure element. It is usually expressed in terms of cube edge a .

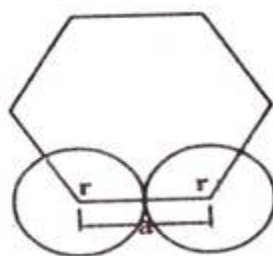


Fig 1.23

To find the atomic radius of the HCP structure, consider any two corner atoms. It has to be noted that, each and every corner atom touches each other; therefore they are the nearest neighbours.

From Figure 1.23, we can write $a = 2r$

$$\text{(or) } r = a/2$$

i.e., the atomic radius is $r = a/2$

Relation between 'c' and 'a' $\left[\frac{c}{a} \text{ ratio}\right]$

Let 'c' be the height of the unit cell and 'a' be the distance between two neighbouring atoms as shown in Fig 1.24 (a). Consider the triangle ABO in the bottom layer of the HCP structure, Fig 1.24 (b). Here A, B and O are the lattice points and exactly above these at a perpendicular distance $c/2$ the next layer atom lies at C, Fig 1.24 (c).

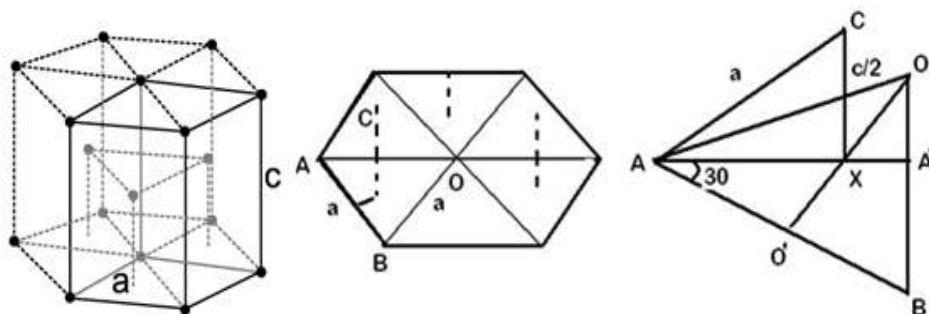


Fig 1.24 (a)**Fig 1.24 (b)****Fig 1.24 (c)**

From the Figure Fig 1.24 (c), in the triangle ABA' $\cos 30^\circ = AA' / AB$

$$\text{Therefore } AA' = AB \cos 30^\circ = a \sqrt{3}/2$$

$$\text{but } AX = 2/3 AA' = 2/3 a \sqrt{3}/2 = a / \sqrt{3}$$

$$\text{In the triangle AXC, } AC^2 = AX^2 + CX^2$$

$$\text{Therefore } CX^2 = AC^2 - AX^2 = a^2 - a^2/3 = 2/3 a^2$$

$$\text{but } CX = c / 2$$

$$\text{Therefore } (c/2)^2 = 2/3 a^2$$

$$\text{(or) } c/a = \sqrt{8/3} = 1.633$$

To find the packing fraction of HCP structure:

Atomic packing factor is defined as the ratio between the volumes occupied by the total number of atoms per unit cell (v) to the total volume of the unit cell (V).

$$\text{i.e., } APF = \frac{\text{Total number of atoms per unit cell} \times \text{Volume of one atom}}{\text{Total volume of the unit cell}} = \frac{(v)}{(V)}$$

$$\text{Area of the base} = 6 \cdot \text{area of the triangle ABO}$$

$$= 6 \cdot 1/2 \cdot AB \cdot OO'$$

$$= 3 \cdot a \cdot \sqrt{3}/2 \cdot a = 3\sqrt{3} a^2 / 2$$

Hence the volume of the unit cell of the HCP structure

$$= \text{Area of the base} \cdot c$$

$$= 3\sqrt{3} a^2 c / 2$$

Considering the top layer which has totally seven atoms, each corner atom contributes 1/6 atom to the unit cell and the centre atom contributes 1/2 atom to the unit cell. Hence the total contribution by the top layer to the unit cell is $1/6 \cdot 6 + 1/2 = 3/2$ atoms. Similarly the bottom layer also contributes the same 3/2 atoms. The central layer contributes 3 atoms fully because all the central layer atoms are well inside the unit cell.

$$\text{Therefore total number of atoms per unit cell} = 3/2 + 3/2 + 3 = 6$$

$$\text{Packing fraction} = \frac{v}{V} = \frac{6 \frac{4}{3} \pi r^3}{V}$$

$$V = 3\sqrt{3} a^2 c/2$$

Substituting $2r = a$ and $c = \sqrt{8/3} a$,

$$\text{We get, Packing fraction} = \frac{v}{V} = \frac{\pi}{3\sqrt{2}} = 0.74$$

Thus the density of packing is 74% and hence it is a close packed structure. Most of the metals crystallize in this structure.

1.13 MILLER INDICES

“The Miller indices are the set of three integers (h k l) used to indicate the different planes in a crystal. The use of Miller indices for notation of crystallographic planes is universal”.

Miller suggested that it is more useful to describe the orientation of a plane by the reciprocal of its numerical parameters rather than by its linear parameters. These reciprocals when appropriately converted to whole numbers are called Miller indices.

Procedure to find Miller indices:

1. Choose x, y, z along crystallographic axis.
2. Find the intercepts of the plane on x, y, z, namely x' , y' , z' .
3. Express x' , y' , z' in terms of the axial units or primitives,
Like, $x' = pa$, $y' = qb$, $z' = rc$.
4. Write the axial units in the form of ratio $p : q : r$ (must be in whole numbers).
5. Find the ratio of their reciprocals $(1/p : 1/q : 1/r)$
6. Convert these reciprocals into whole numbers by multiplying each with their LCM to get smallest whole number. This gives the Miller indices (h k l) of the given plane.

In Fig.1.25 this plane intercepts the axes at $3a$, $2b$, and $2c$.

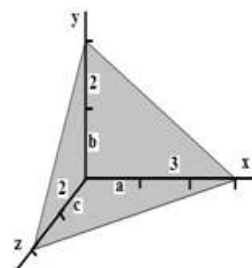


Fig.1.25

The reciprocals of these numbers are $\frac{1}{3} : \frac{1}{2} : \frac{1}{2}$

The smallest three integers having the same ratio (when multiply by 6) are

2, 3, 3, and thus the Miller indices of the plane are $(h\ k\ l) = (233)$.

The same $(h\ k\ l) = (233)$ will be the values for the parallel planes.

Braces $\{h\ k\ l\}$ signify all the planes that are equivalent in a crystal.

For example in a cubic crystal $\{100\}$ is equal to the set its six faces (100) , (010) , (001) , $(\bar{1}00)$, $(0\bar{1}0)$, $(00\bar{1})$.

The symbol for a family of parallel planes is $\langle hkl \rangle$.

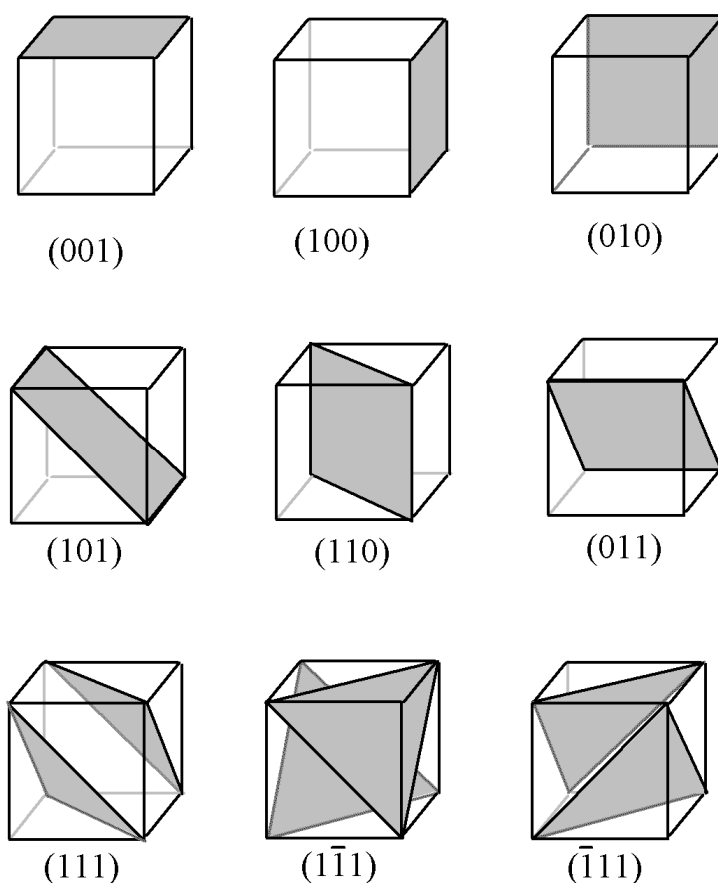


Fig1.26

1.14 X-RAY DIFFRACTION AND BRAGG'S LAW

When a beam of fast moving electrons strike a solid target an invisible high penetrating radiation is produced. These high penetrating radiations are called X-rays. Actually X-rays are electromagnetic waves like visible light, but of very short waves length. W.L. Bragg discovered that cleavage planes of crystals could reflect X-rays when the X-rays are incident on their surface nearly

at glancing angles. Glancing angle or Bragg angle is the angle between incident ray and crystal plane.

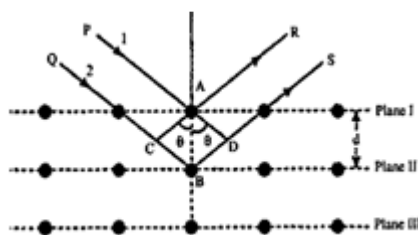


Fig 1.29

Theory

Figure 1.29 shows a beam of monochromatic X-rays incident at a glancing angle ' θ ' on a set of parallel planes of a crystal. The beam is partially reflected at the successive layers rich in atoms. Ray no. 1 is reflected from atom A in plane 1 whereas ray no. 2 is reflected from atom B lying in plane II immediately below atom A. Whether these two reflected rays will be in phase with each other will depend on their path difference.

This path difference can be found by drawing perpendiculars AC and AD on ray no. 2. Since the two rays travel the same distance from A and D onwards, it is obvious that ray no. 2 travels an extra distance = CB + BD.

Hence the path difference between the two reflected beams = CB + BD

$$= d \sin \theta + d \sin \theta = 2 d \sin \theta$$

Where ' d ' is the interplanar spacing i.e., vertical distance between two adjacent planes belonging to the same set.

The two reflected beams will be in phase with each other if this path difference equals an integral multiple of λ and will be out of phase if it equals an odd multiple of $\frac{\lambda}{2}$. Hence the condition for producing maxima (or) constructive interference becomes,

$$2 d \sin \theta = n\lambda$$

Where $n = 1, 2, 3, \dots$ For the first order, second order and third order maxima respectively. This relation is known as Bragg's law.

1.15 METHODS OF DETERMINATION OF CRYSTAL STRUCTURE

The important methods of determining crystal structure are

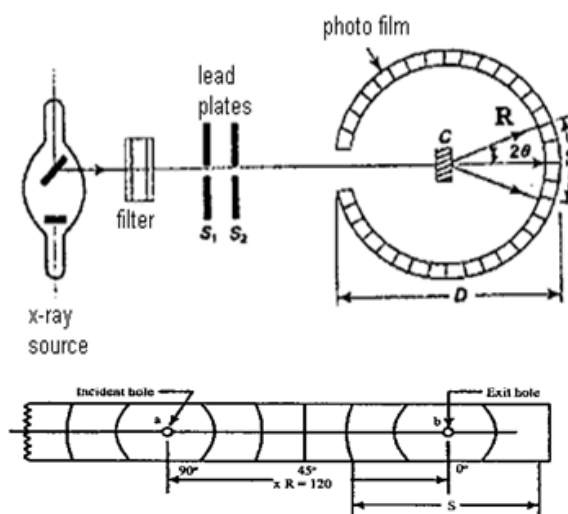
- (i) X-ray diffraction method

- (ii) Electron diffraction method
- (iii) Neutron diffraction method

Powder crystal Method/Debye – Scherrer Method

The powder method is applicable to finely divided crystalline powder or to a very fine grained polycrystalline specimen. One form of powder photographic is invented by P. Debye and Scherrer so this method is also known as Debye – Scherrer method. The powder method is more convenient since single crystals are not required and it is used for accurate determination of lattice parameters (a) in crystal of known structure and for the identification of elements and compounds. The fine grain powder is compressed into a thin rod or packed into a small capillary tube. A narrow beam of monochromatic X-rays is made to fall on the powder and recorded by the photographic film as a series of lines of varying curvatures as shown in Fig.1.30.

The full opening angle of the diffraction cone 4θ is determined by measuring the distance ‘S’ between two corresponding arcs on the powder photograph symmetrically displayed about the exit point of the direct beam.



Flattened Photographic film after developing and indexing of different lines

Fig 1.30

The distance ‘S’ on the film between two diffraction lines corresponding to a particular plane is related to the Bragg angle by the equation

$$4\theta = \frac{S}{R} \text{ radians}$$

$$4\theta = \frac{S}{R} \left(\frac{180}{\pi} \right) \text{ degrees}$$

Where 'R' is the specimen to film distance, usually the radius of the camera housing the film. A list of 'θ' values can be obtained from the measured values of S. Since we get number of lines it is marked as S1, S2, S3... etc and the corresponding $\theta_1, \theta_2, \theta_3, \dots$ values can be obtained. Since the wavelength is known i.e., wavelength of the copper (anode) target from which we produce X – rays its wavelength is $\lambda = 1.5418\text{\AA}$. By substituting the values of θ and λ in the Bragg's formula $2d \sin \theta = n\lambda$ (where $n = 1$ for first order) we get a list of 'd' spacing.

- (i) If we know 'a' the lattice constant $\left[a = \left(\frac{nN}{N\rho} \right)^{\frac{1}{3}} \right]$ we can also calculate (generate) Miller indices (h k l) by substituting the 'a' and 'd' values in

$$d_{(hkl)} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- (ii) If we know Miller indices (h k l) we can determine the structure of the crystal.

Advantages of the Powder Crystal Technique

The powder technique has the advantage that it does not require large single crystal and almost any substance can be ground into powder. By measuring the distance between symmetrical lines (S) and the lines intensities, following information can be obtained.

- A crystalline substance can be distinguished from an amorphous one because the former produces line on a photograph, the latter does not.
- If photographs are made before and after heat treatment of a crystalline substance, their comparison can reveal any alteration in the lattice.
- With the aid of standard photographs of known chemical compounds, their presence may be detected in an unknown mixture.
- The size of the unit cell and the type of the lattice can be determined in simple cases.

1.16 CRYSTAL DEFECTS

So far we have seen ideal crystals obtained by combining a basis with a infinite space lattice. Now let us consider real crystals. Since the real crystals are of finite size, they have a surface as their boundaries. The boundary atomic bonds terminate and hence the surface itself is an imperfection. In addition there is other occasional disruption in periodicity within a crystal. Many of the properties of crystalline materials are structure sensitive properties. They mainly depend on the presences or absences of imperfection can be classified on the basis of their geometry as follows,

1. Point defects
2. Line defects
3. Surface defects

4. Volume defects.

1.17 POINT DEFECTS

The defects which take place due to imperfect packing of atoms during crystallization are known as point defects.

As the name implies, they are imperfect point like regions in the crystal and, therefore, they are also referred as Zero dimensional imperfections.

The most common point defects in a crystal are described below,

- (a) Vacancies
- (b) Substitutional impurity
- (c) Interstitial defects
- (d) Self-interstitial defects
- (e). Frenkel defect
- (f). Schottky defect

(a) Vacancies

A vacancy is a point defect which occurs due to the missing of an atomic site which creates a vacant as shown in Fig. 1.31. This vacancy occurs due to thermal vibration of atoms at high temperature.

(b) Substitutional impurity

A substitutional impurity defect is a point imperfection which refers to a foreign atom that substitute for or replace a parent atom in the crystal as shown in Fig 1.31.

Pentavalent or trivalent impurity atoms doped in Germanium or Silicon are also substitutional impurity.

(c) Interstitial defects

Interstitial impurity is created when impurity atom occupies the interstitial position (void) without replacing the parent atom in the crystal. These can exist only in ionic and covalent crystals. Here the impurity atom has very small size as that of the host atoms. The presence of carbon in iron is an example for this defect as shown in Fig 1.31.

(d). Self-interstitial defects

It occurs due to the occupation of same atom in the interstitial position of the crystal lattices.

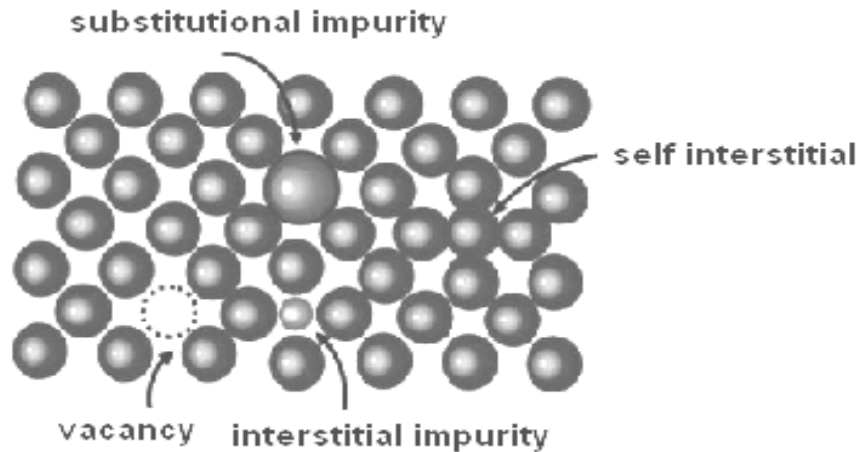
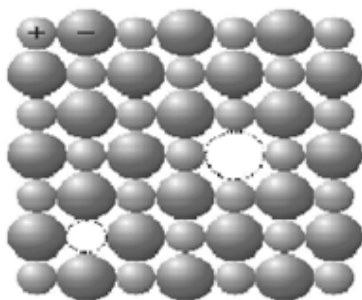


Fig 1.31

(e). Schottky defect

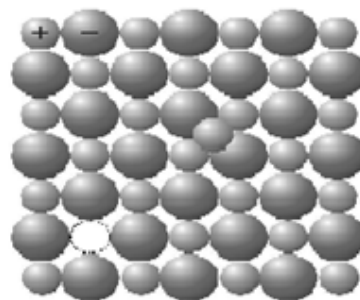
Schottky vacancies refer to the missing of anion and cation as shown in Fig.1.32(a) . In general the missing of pair of ions in ionic crystal is called schottky defect.

The Schottky vacancies are usually produced as a result of the thermal incorporation of the unoccupied lattice sites from the exterior of the crystal. The lattice undergoes thermal vibration and thermal expansion when temperature is raised above 0 K. When it happens, a pair of vacancies is incorporated into the crystal, probably near the surface. This may further diffuse the interior of the crystal. This shows that the vacancies may be increased by increasing temperature.



schottky defect

Fig 1.32 (a)



frenkel defect

Fig 1.32 (b)

(f). Frenkel defect

Frankel vacancies refer to the shift of cation from the regular site to interstitial site as shown in Figure 1.32(b). Such vacancies are produced when crystal is exposed to high radiation. It always occurs in ionic crystals. Normally the cation is shifted from the regular site to the interstitial site. This because the cation being very small, it can be easily accommodated in the void space.

Here, the radiation knock some atoms out of their normal lattice positions and force them into the interstices of the structure, thus creating simultaneously vacancies and interstitials.

It is also a temperature dependent, i.e., higher is the vacancy, and higher is the temperature.

1.18 LINE DEFECTS (OR) DISLOCATIONS

The defects which take place due to dislocations or distortion of atoms along a line in some direction is called line defect.

The dislocation is a boundary between the slipped region and the unslipped region and lies in the slip plane. The structure and behaviour of dislocations affect many of the properties of engineering materials.

This is nothing but one part of the crystal shifts or slips relative to the rest of the crystal. The imperfectionals are formed due to solidification of metals.

Dislocations are generally classified into four types. They are (a) edge dislocations (b) screw dislocation (c) dislocation slip and (d) dislocation climb

(a) EDGE DISLOCATION

An edge dislocation arises when one of the atomic planes forms only partially and does not extend throughout the entire crystal.

The atomic planes abruptly terminate at centre of shaded circle. It is viewed as an extra plane inserted in between a set of parallel planes shown in Fig. 1.33.

The atomic row 5 passing through point centre has one atom more than the row adjacent to it. The edge of such a plane forms a line defect and it is called an edge dislocation.

Edge dislocation is symbolically represented as T and $\perp$, positive edge dislocation and negative edge dislocation.

Positive edge dislocation:

If extra plane of atoms is above the slip plane of the crystal, then the edge dislocation is called as positive edge dislocation ($\perp$)

Negative edge dislocation:

If the extra plane of atoms is below the slip planes then the dege dislocation is called negative edge dislocation (T)

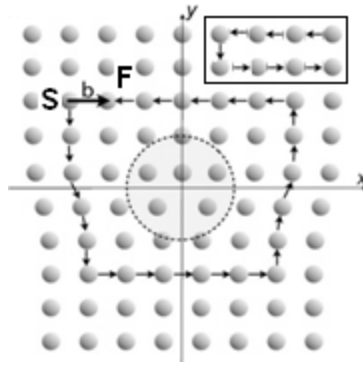


Fig 1.33

BURGER'S VECTOR

The dislocation lines can also be described by Burger's vector (b). The vector which indicated the direction and magnitude of the shift of the lattice on the slip plane is called a Burger vector.

Burger vector of a dislocation line can be found by using Burger's loop (or) circuit method. Burger's circuit or loop is formed by taking integral lattice translations in each pair of parallel sides in the region of interest. Such a circuit would close itself in a perfect region of the crystal a region with no dislocation will form a closed circuit. If a Burger circuit is described around the dislocation position of the lattice, the start and finish will not be coincident as shown in Fig. 1.33.

If a Burger circuit is described around the dislocation position of the lattice, the start and finish will not be coincident. Let us consider a point S and move down by 5 steps and to the left by 5 it ends up at the point F instead of ending at S. Now, we have to move an extra step $\overrightarrow{FS}$ to return to the starting point S in order to close the circuit. The magnitude and direction of the extra step defines the Burger's vector ($\overrightarrow{b}$).

(b) Screw dislocation

Screw dislocation is due to a displacement of atoms in one part of a crystal relative to rest of the crystal. The displacement terminates within the crystal. This dislocation forms a spiral ramp around the dislocation line as shown if Fig. 1.34 .

In a screw dislocation there is a line of atoms about which the crystal planes are twisted to give an effect similar to threads of a screw.

The row of atoms marking the termination of a displacement is the screw dislocation .The term screw represents that one part of the crystal is moving in spiral manner about the dislocation line

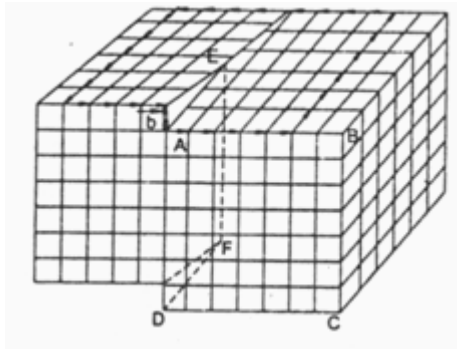


Fig 1.34

1.19 SURFACE DEFECTS (OR) PLANE DEFECTS (OR) TWO DIMENSIONAL DEFECTS

The defects, which take place on the surface of a material, are known as surface defects or plane defects. The surface defects take place either due to imperfect packing of atoms during crystallization or defective orientation of the surface.

The following are the different types of surface defects.

(a) Grain boundary

Grain boundaries are the defects which separate grains of different orientation from each other in a polycrystalline material.

Grain boundary is formed when two growing grain surfaces meet. Figure 1.35 shows a two dimensional analogy of the grain boundary. Grain boundaries are observed during the solidification of polycrystalline material. Grain boundary area depends on the grain size of the material and increases with decrease in grain size and vice versa.

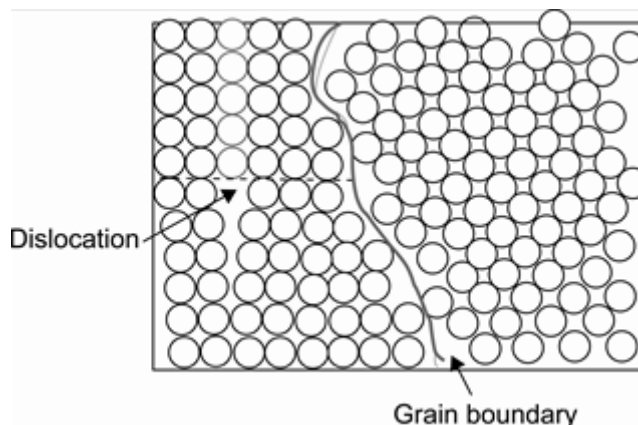


Fig 1.35

(b) Twin boundary

These are the boundaries in the grains at which the atomic arrangement on one side of the boundary is the mirror image of the atoms on the other side. The twin boundary is given below is illustrated in Fig. 1.36.

The volume of material which has an orientation similar to the mirror image of the matrix operation is called a twin. The mirror is called the twinning plane (or) composition plane.

Twins may be formed during solidification, deformation or during deformation and annealing of metals.

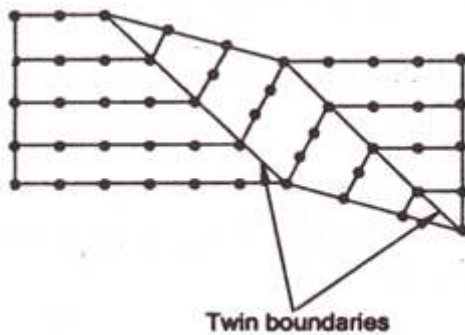


Fig 1.36

(C). Tilt boundary

Tilt boundary is another type of surface defect. It is called a small angle boundary as the orientation of grains in two neighbouring crystals differs by only a few degrees. i.e., less than 10° . A low angle tilt boundary consists of edge dislocation lying one above the other in the boundary and is associated with relatively little energy in the below Fig. 1.37.

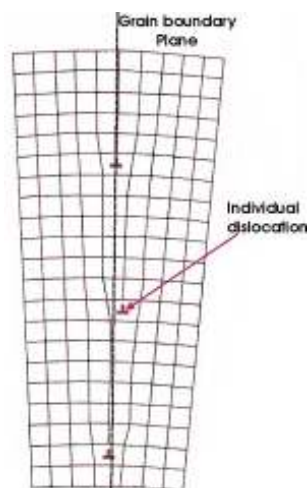


Fig 1.37

The Fig. 1.37 shows a simple tilt boundary composed of edge dislocation at distance “h” apart in a simple cubic lattice.

Therefore, the angle of tilt will be $\theta = \frac{b}{h}$, where “b” is the magnitude of the Burger’s vector and “h” is the average vertical distance between the dislocations.

These defects affect the deformation behaviour, mechanical properties and recrystallisation temperature of materials. They also have an influence on electrical properties and corrosion resistance.

(c) Stacking fault

Stacking faults are planar surface imperfections, and, are caused by fault in the stacking sequence of atomic planes in crystals of FCC and HCP materials.

Fig. 1.38 shows the proper sequence of atomic planes ABC ABC ABC.... in a FCC structure if we read from bottom to top. Fig. 1.39 shows the sequence of atomic planes ABC ABA BABC.... the region in which stacking fault occurs is AB AB. This thin region is a surface imperfection and is called a stacking

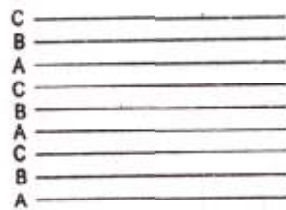


Fig.

1.38

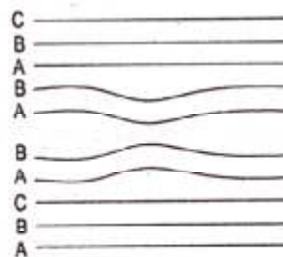


Fig.

1.39

1.20 VOLUME DEFECTS (OR) THREE DIMENSIONAL DEFECTS (OR) BULK DEFECTS

Volume defects such as cracks may arise when there is only small electrostatic dissimilarity between the stacking sequences of close packed planes in metals. In addition, whenever a cluster as missing, large vacancy (or) void is got. This is also a volume defect.

Mosaic structure and inclusions are the defects which are distributed volume wise.

Foreign particle inclusions, large voids or non-crystalline regions which have the dimensions of the order of 20 are also called volume imperfections.

Problems:

1. A crystal with face centered cubic lattice has density 6250 kg/m^3 and molecular weight 60.2. Calculate the lattice constant.

Solution :

$$\text{Given } \rho = 6250 \text{ kg/m}^3$$

$$A = 60.2$$

$$n = 4$$

$$a = ?$$

Formula

$$\rho = \frac{nA}{Na^3}$$

$$(\text{or}) a^3 = \frac{nA}{\rho N} = \frac{4 \times 60.2}{6.023 \times 10^{26} \times 6250} = \frac{240.8}{3.764375 \times 10^{30}}$$

$$(\text{or}) a^3 = 6.3968 \times 10^{-29} \text{ m}^3$$

$$\therefore a = 4.007 \times 10^{-10} \text{ m}$$

2. Lattice constant of copper is 0.38 nm. Calculate the distance between (110) planes.

Solution:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$d = \frac{0.38 \times 10^{-9}}{(1^2 + 1^2 + 0^2)^{1/2}}$$

$$d = \frac{0.38 \times 10^{-9}}{\sqrt{2}}$$

$$d = 0.27 \times 10^{-9}$$

$$d = 0.27 \text{ nm}$$

3. A beam of X-rays are incident on a sodium chloride crystal (lattice spacing 0.282 nm). The first order Bragg reflection is observed at a glancing angle of $8^\circ 35'$. What is the wavelength of X-rays?

Solution:

$$\text{Bragg's law : } 2d \sin \theta = n\lambda$$

$$\text{Given } d = 0.282 \text{ nm}$$

$$\theta = 8^\circ 35'$$

$$n = 1$$

Hence

$$\lambda = \frac{2d \sin \theta}{n}$$

$$\lambda = \frac{2 \times 0.282 \times 10^{-9} \times \sin 8^\circ 35'}{1} = 0.0842 \text{ nm}$$

4. X-rays of wavelength 1.5418 Å are diffracted by (111) planes in a crystal at an angle 30° in the first order. Calculate the interatomic spacing.

Solution:

Given

$$\lambda = 1.5418 \text{ Å}$$

$$\theta = 30^\circ$$

$$n = 1$$

Bragg's law; $2d \sin \theta = n\lambda$

$$\therefore d = \frac{n\lambda}{2 \sin \theta} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\text{Given } \lambda = 1.5418 \times 10^{-10} \text{ m}$$

$$\theta = 30^\circ$$

$$n = 1 (\text{first order})$$

$$d = \frac{1.5418 \times 10^{-10}}{2 \times \sin 30^\circ} = 1.5418 \times 10^{-10}$$

$$a = \sqrt{h^2 + k^2 + l^2}$$

$$= 1.5418 \times 10^{-10} (1^2 + 1^2 + 1^2)^{1/2}$$

$$= 1.5418 \times 10^{-10} \times 1.732 = 2.670 \text{ Å}$$

5. Zinc has HCP structure . If the lattice constant is 3.2Å, find the volume of unit cell?

Volume = 6Xarea of triangle X C (Jan 2015)

$$= 6 \times \frac{1}{2} a^2 \frac{\sqrt{3}}{2} \times \frac{\sqrt{8}}{\sqrt{3}} a$$

$$= \frac{3}{2} a^3 \sqrt{3} (1.633)$$

$$= \frac{3}{2} (0.32)^3 \sqrt{3} (1.633) = 1.39 \times 10^{-28} \text{ m}^3$$

Pondicherry University

Part-A Question and Answer

1. Define space lattice (or) crystal lattice. (April 2010)

A three dimensional collection of point in space is called a space lattice (or) crystal lattice. The environment about any particular point is in every way the same.

2. Define unit cell. (Nov 2011)(Jan2013)(Jan 2011)

The unit cell is defined as the smallest geometric Figure, the repetition of which gives the actual crystal structure.

(or)

A unit cell in three dimension is defined as that volume of the solid, from which the entire crystal can be formed by the translational repetition in three dimension.

3. What are Bravias lattices? (May 2012, 2014)

The 14 possible ways of arranging points in space lattice such that all the lattice points have exactly the same surroundings. These 14 lattices are called the Bravais lattices.

4. Define coordination number. (Jan 2015) (April 2011) (Jan 2011)

It is the number of nearest neighbouring atoms to a particular atom.

(or)

Coordination number is the number of nearest neighbours directly surrounding a given atom.

5. Define Burger vector.(Jan 2014) (Nov 2011)

The vector which indicates the direction and magnitude of the shift of the lattice on the slip plane is called Burger vector.

6. Bismuth has $a=b=c=4.74\text{Å}$ and angles $\alpha=\beta=\gamma$ not equal 60° . What is its crystal structure? (May 2012)

The crystal structure is Rhombohedral. Since $a=b=c$ and $\alpha=\beta=\gamma$ not equal 90°

7. Name seven crystal systems. (April 2011)

Cubic, Tetragonal, Orthorhombic, Rhombohedral, Hexagonal, Monoclinic and Triclinic.

Additional Part-A Question and Answer

1. What are Miller indices?

Miller indices are the smallest possible integers which have the same ratios as the reciprocals of the intercepts of the plane concerned on the three axis.

2. What is a primitive cell?

A primitive cell is the simplest type of unit cell which contains one lattice point per unit cell.

3. Name the crystal structure of the following:

- (a) Gold – FCC
- (b) Germanium – Diamond cubic
- (c) Barium – BCC
- (d) Zinc – HCP

All the above elements exhibit the structures given above at 20°C. But for Barium other structures also exists at other temperatures.

4. What is a crystal? What are crystalline materials? Give example.

Crystalline solids (or) crystals are those, in which the constituent atoms (or) molecules are arranged in an orderly fashion throughout in a three dimensional pattern. Example: Copper, Silver.

5. What is an amorphous solid? Give example.

It is type of solid, in which the atoms (or) molecules are not arranged in an orderly fashion. i.e., The same atomic groups are arranged more randomly.

Example: Plastic, rubber

6. What is a Basis (or) Motif?

A unit assembly of atoms (or) molecules (i.e., one or more atoms (or) molecules) identical in composition, arrangement and orientation is called the Basis (or) Motif.

A crystal structure is formed by associating a Basis to every lattice point in a space lattice.

7. What is a crystal structure?

A crystal structure is formed by associating a basis to every lattice point in a space lattice.

i.e., crystal lattice Basis + space lattice

8. What are lattice parameters?

The intercepts a , b and c of a unit cell and the interfacial angles α , β and γ along three axes are called the lattice parameters.

9. Define atomic radius.

It is defined as half of the distance between two nearest neighbours in a crystal of pure element

10. What are point defects? (or) Zero dimensional defects?

The defects which take place due to imperfect packing of atoms during crystallisation are known as point defects.

11. What are Frenkel and Schottky defects?

Frenkel defect is an ionic crystal imperfection that occurs when an ion moves in to an interstitial site, thereby creating two defects simultaneously i.e., one vacancy and the other self interstitial.

A pair of ion vacancies in an ionic crystal is termed as Schottky defect.

12. What is a line defect (or) a one dimensional defect?

The defects which take place due to dislocation or distortion of atoms along a line in some direction is called line defect.

13. What is edge dislocation?

It is a region of lattice disturbance extending along an edge inside a crystal due to the insertion of an extra plane of atoms.

14. What is screw dislocation?

Screw dislocation results from a displacement of the atoms in one part of a crystal relative to the rest of crystal forming a spiral ramp around the dislocation line.

15. What is meant by stacking fault?

The stacking faults are planer surfaces imperfection and, are caused by faults as the stacking sequence of atomic planes in crystals of FCC and HCP materials.

16. What are the differences between edge and screw dislocation?

Sl.No.	Edge dislocation	Screw dislocation
1.	These dislocations arise due to introduction or elimination of an extra plane of atoms.	Screw dislocation results from a displacement of the atoms in one part of a crystal relative to the rest of crystal forming a spiral ramp around the dislocation line.

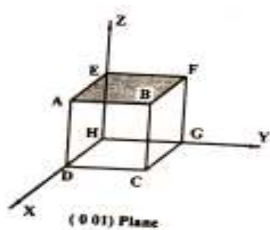
2.	Region of lattice disturbance extends along an edge inside a crystal	Region of lattice disturbance extends in two separate planes at right angles to each other.
3.	An edge dislocation can glide and climb	A screw dislocation can glide only.
4.	Burger vector is always perpendicular to the dislocation line.	Burger vector is parallel to the dislocation line.
5.	These are formed during deformation and crystallization.	These are also formed during deformation and crystallization.

17. What is meant by polymorphism and allotropy?

Polymorphism: A substance that can exist in two or more forms in the same state is called polymorphism.

Allotropy: An element that can exist in two or more forms in the same state is called allotropy.

18. Draw the crystal planes with the Miller Indices (001).



19. What is crystallographic axis? (April 2010)

The lines drawn parallel to the lines of intersection of any three faces of the unit cell which do not lie in the same plane are called crystallographic axis. In Cubic Crystal System, the three crystal axes are perpendicular to each other and the axial length is the same along the entire three axes.

20. State the condition imposed on the cell parameters for crystal systems having the largest number of Bravais lattices and the least number of nearest neighbours.

- Crystal with least number of nearest neighbours is simple. (6 neighbours).
- Crystal system with large number of Bravais lattices is orthorhombic, (4 Bravais lattices)

Pondicherry University
Part-B Question and Answer

1. Determine Coordination Number, Atomic radius, c/a ratio and Packing Factor for HCP structure. **(May 2014)(Jan 2014)(April 2010)**
 - (a) What are Miller Indices? Explain how they are determined. **(Jan 2014)**
 - (b) Explain point defects with appropriate diagrams.
2. Determine No. of atoms per unit cell, Atomic Radius, Coordination Number and Packing Factor for FCC structures. **(April 2011)**
3. Explain in detail the point, line and surface defects of the crystal with appropriate diagrams.
4. (a) Show that hcp structure demands an axial (c/a) ratio of 1.633.
(May 2012)
 - (b) Determine atomic radius, coordination number and packing factor for FCC structure. **(Jan 2013)**
5. (a) What is meant by point defects in crystal lattice and Explain in detail how these point defects are caused? **(Nov 2011)**
 - (b) Explain screw and edge dislocation of crystal imperfections.
6. Prove that the packing factor of HCP is 74%. **(Nov 2011)**
7. Determine the atomic radius, Co-ordination number and atomic packing factor of FCC and HCP structure.
8. What are point and line imperfections in solid materials? Illustrate these imperfections with suitable sketches **(May 2012)**
9. Explain in detail the point, line and surface defects of the crystals with appropriate diagrams?
(Jan 2013) (April 2011)
10. Write notes on **(April 2010)**
 - (i) Coordination number (ii) Atomic radius (iii) Explain edge dislocation and screw dislocation